MTH 335(A) Dr. James Ashe
Homework 2 Due date: March 4, 2013 **Name:**

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.*

1. Prove or disprove the following.

a. A constant function is not one-to-one.

False. Let $A = \{a\}$ and $f : A \to B$ and $f(a) = b$ for all $a \in A$. If $f(a_1) = f(a_2)$ then $a_1 = a_2$ because A only has one element. So f is constant and also one-to-one. *In general constant functions are not 1-1, i.e., the statement "constant functions are 1-1" is false.* $\square$

b. The identity function is one-to-one.

True. Let $f : A \to A$ be the identity function. If $f(a_1) = f(a_2)$ then $f(a_1) = a_1 = a_2 = f(a_2)$ because $f(a) = a$ for any $a \in A$. Thus f is one-to-one. $\square$

c. The identity function is onto.

True. Let $f : A \to A$ be the identity function and $a \in A$. $f(a) = a$ so then f is onto.

$\square$

2. Let $g : A \to B$ and $f : B \to C$. Prove that if $f \circ g : A \to C$ is one-to-one and g is onto then f is also one-to-one.

Proof. Let $b_1, b_2 \in B$ such that $b_1 \neq b_2$. Because g is onto there is a $a_1, a_2 \in A$ such that $g(a_1) = b_1$ and $g(a_2) = b_2$. So then $f(b_1) \neq f(b_2)$ since $f(g(a_1)) \neq f(g(a_2))$.

$\square$

3. Suppose that $f : A \to B$ is a one-to-one function. Prove or disprove that if $f^{-1} : f(B) \to A$ is a function then f^{-1} is also one-to-one.

True. Recall: It was shown in class that for a function g, the inverse of g is a function if and only if g is $1-1$. Since f^{-1} is a function and $(f^{-1})^{-1} = f$ is also a function, it follows that f^{-1} is $1-1$. $\square$

Alternatively,

Proof. Suppose to the contrary that f^{-1} is *not* $1-1$. So then there is $b_1, b_2 \in B$ such that $b_1 \neq b_2$ and $f^{-1}(b_1) = f^{-1}(b_2) = a$ for some $a \in A$. Thus $f(f^{-1}(b_1)) = b_1 \neq b_2 = f(f^{-1}(b_2))$ which contradicts the assumption that f is a fucntion. $\square$

4. Let $f : A \to B$ be a one-to-one function. Show that $|A| \leq |B|$.

Proof. Let $|A|$ be the number of items and $|B|$ be the number of boxes. By the pigeon hole theorem, if $|A| > |B|$ then f would map (or place) at least 2 items to the same box and thus not be $1-1$. Thus f must be $1-1$.

$\square$

5. Let $f : A \to B$ and $g : B \to C$ be functions. Prove that $g \circ f : A \to C$ is a function.

Proof. f is a funciton implies that for any $a \in A$ there is a unique $b \in B$ such that $f(a) = b$. Also, g is a function implies that for any $b \in B$ there is a unique $c \in C$ such that $g(b) = c$. Thus for any for any $a \in A$ there is a unique $c \in C$ such that $g(f(a)) = g(b) = c$. So $g \circ f$ is a fucntion. □

6. (This problem was dropped) Let $f : A \to A$, $g : A \to A$, and $h : A \to A$; prove that $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. Let $a_0 \in A$. Since f, g, h are functions there are $a_1, a_2, a_3 \in A$ such that, $f(a_0) = a_1$, $g(a_1) = a_2$, and $h(a_2) = a_3$. $((h \circ g) \circ f)(a_0) = h(g(f(a_0))) = h(g(a_1)) = h(a_2) = a_3$. Also, $(h \circ (g \circ f))(a_0) = h((g \circ f)(a_0)) = h(g(f(a_0))) = h(g(a_1)) = h(a_2) = a_3$. Thus $(h \circ g) \circ f = h \circ (g \circ f)$. □

7. Prove that if $f : A \to B$ is onto and $g : B \to C$ is onto, then $(g \circ f) : A \to C$ is onto.

Proof. Let $c \in C$. g is onto so then there exists a $b \in B$ such that $g(b) = c$. Since f is onto there also exists an $a \in A$ such that $f(a) = b$. So then $g(f(a)) = g(b) = c$; thus $g \circ f$ is onto. □

8. Each of the following statements, $P(x, y)$, define a relation R in the natural numbers, $\mathbb{N}$, i.e., $R = (\mathbb{N}, \mathbb{N}, P(x, y))$. Show whether each of the following relations is reflexive.

a. $P(x, y) =$"x is less than or equal to y"

Reflexive: Let $x \in \mathbb{N}$. $x = x \Rightarrow x \leq x$ so then R is reflexive. □

b. $P(x, y) =$"x divides y"

Reflexive: Let $x \in \mathbb{N}$. $x = 1 \cdot x$ so then $x|x$. Thus R is reflexive. □

c. $P(x, y) =$"$x + y = 10$"

Not Reflexive: Consider the natural number 4; $4 + 4 \neq 10$. Thus R is not reflexive. □

d. $P(x, y) =$ "x and y are relatively prime."

Not Reflexive: Consider the natural number 8; $\gcd(8, 8) = 8 \neq 1$ since $8|8$. Thus R is not reflexive. □

9. Extra Credit: Did you attend the STEM college Lyceum given by Dr. Stephenson?

Proof. YES of course. It was better than a Beyonce concert at Disney World. □

Presentation problems.

1. Let the function $f : A \to B$ be 1-1 and onto, i.e., f^{-1} exists and $|A| = |B|$. Prove that $\left(f^{-1} \circ f\right) : A \to A$ is the identity function on A and that $\left(f \circ f^{-1}\right) : B \to B$ is the identity function on B.

2. Let $f : A \to B$ be 1-1 and onto. Prove that $g = f^{-1}$, if $(g \circ f) : A \to A$ is the identity function on A and $(f \circ g) : B \to B$ is the identity function on B.

3. The *product set* of sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$; it is denoted by $A \times B$. Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

4. Prove: Let R and R' be symmetric relations in a set A; then $R \cap R'$ is a symmetric relation in A.

5. Prove that $\displaystyle\sum_{k=1}^{n} 2k = n^2 + n$.

6. The Fibonacci numbers are :$1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$. In general, the Fibonacci numbers are defined by $f_1 = 1$, $f_2 = 1$, and $f_{n+2} = f_{n+1} + f_n$ for $n = 1, 2, 3, \ldots$. Prove that the n^{th} Fibonacci number $f_n < 2^n$.

7. Prove that there infinitely many prime numbers.

☺. Two sets are said to be *equal* if there exists a bijective function between them; we write $|A| = |B|$. B is said to have *strictly greater cardinality* than B if there is a one-to-one function from A into B, but there does not exist a bijective function from A into B; we write$|A| < |B|$. Prove that $|\mathbb{N}| < |\mathbb{R}|$.